

J-Lens in Ouro: Early-Pass Deficits and a Confirmed Late-Pass Advantage

Jan Kirin

Working draft v2 · 11 September 2026

Abstract

Does a Jacobian-based readout reveal intermediate content that a model’s own output head misses during recurrent computation? I compare J-Lens with the raw logit lens in frozen Ouro-2.6B, which applies 48 shared blocks over four recurrent passes. On multihop questions the raw lens was stronger in passes 1–3, while layerwise analysis found a positive J-Lens region in pass 4 whose location and size five independent calibration fits reproduced on the discovery set. I then froze one estimator (fit01) and the band, physical layers 26–37 of pass 4, before evaluating 160 new two-hop questions with 80 new intermediate concepts. In that band J-Lens improves excess hit@10 by **23.19 percentage points** (95% dependency-group bootstrap interval **16.29–32.90**); intended-concept recovery is 37.92% against 14.43% for the raw lens. The same physical band shows the opposite sign in passes 1–3 (−6.5, −8.6 and −11.1 points; post-confirmation, simultaneous intervals excluding zero), so the pattern is pass-dependent at fixed depth. Replacing the 79 controls by the seven same-domain names lowers the gain to 20.8 points (9.1–32.5), because the lens also recovers same-domain names about nine times more often than arbitrary ones. An exhaustive tokenized-input audit finds no intended-label token in any prompt and six control-token mentions whose removal moves the estimate by at most 0.3 points. Changing the derivative target or the position aggregation substantially reduces the late advantage. FP64 reference scoring and a cross-hardware state replay change the primary estimate by at most 0.42 points. On the discovery set, retained readouts of application-era banks that target each pass’s own exit recover more early-pass content than both the final-target lens and the raw lens; the corresponding test on the confirmation set is blocked because those banks survive only in a remote repository. The result supports a localized, estimator- and target-specific improvement in concept recovery, not general J-Lens superiority, improved answer accuracy, or causal use of the recovered content.

1. The question

A readout can miss useful content even when that content is present in a model’s hidden state. The raw logit lens applies the model’s output head directly to an intermediate activation. J-Lens instead first applies a fixed, Jacobian-derived map that averages how activations affect downstream states across calibration contexts, rather than fitting a predictor to one task’s answers. Its intended target is verbalizable content, not accurate next-token prediction in every context [1].

Ouro gives this comparison two depth axes: physical layer and recurrent pass. The same layer stack is reused, so a readout that works at one pass need not work at another. The experiment asks where each method recovers a task-annotated intermediate concept more reliably. It does not assume that a recovered word was part of the model’s actual reasoning [2].

My initial expectation was that J-Lens would provide a broadly better view of intermediate computation. That expectation did not survive the early-pass results. The positive finding came later: a specific region of the final pass where J-Lens outperformed raw readout. This paper separates the exploration that found that region from the new-question test that checked it, and reports the

post-confirmation reviewer analyses separately from both.

The distinction matters beyond this comparison. Hidden-state readouts could complement observation of visible reasoning, but only after establishing what their scores measure. No monitoring system, deception detector, or chain-of-thought baseline is evaluated here.

2. Readouts and measurement

2.1 Model and fixed estimators

The model is **ByteDance/Ouro-2.6B**, revision 1ed04250da1a9936042725d302e81c8fa2ab5abd (not the Thinking or RLTT checkpoint), evaluated frozen in BF16 with four passes through 48 blocks, residual width 2,048 and a vocabulary of 49,152 tokens. I use one-based physical layers ℓ and pass numbers r ; the stored zero-based virtual index is $v = 48(r-1) + (\ell-1)$. Ouro re-normalizes the residual stream at the end of every pass before carrying it into the next pass and before the shared output head, so a recorded state at the last block of a pass is the pre-normalization block output.

For a captured residual state $h_{i,r,\ell}$ at the readout position of item i , let N be the model’s final RMS normalization and W_U its unembedding. The two vocabulary readouts are

$$z_{i,r,\ell}^{\text{raw}} = W_U N(h_{i,r,\ell}), \quad z_{i,r,\ell}^J = W_U N(B_{r,\ell} h_{i,r,\ell}),$$

where $B_{r,\ell}$ is the frozen J-Lens matrix for that virtual layer. Ranking the logits gives the token readout; no task classifier is trained on the confirmation questions. The main banks target the final residual state of pass 4 (virtual layer 191, before the final normalization). The original J-Lens study’s default Sonnet configuration targets the penultimate layer, so this is an adaptation, not an exact reproduction of that configuration [1].

Fitting recipe (from the executed records). Each bank averages, over 100 calibration paragraphs, the Jacobian estimator of the released implementation: for every source layer v and output dimension, a one-hot cotangent is placed at every valid target position of the target layer at once and back-propagated through the BF16 model with ordinary autograd, giving at source position p the sum over later positions $t' \geq p$ of $\partial h_{191}[t'] / \partial h_v[p]$; the mean over valid source positions (positions 16 to len−2 of a 128-token input, BOS included) is the paragraph’s matrix. Matrices are accumulated as an equal-paragraph FP32 sum, divided by 100 once and stored in FP16; at readout they are cast back to FP32. Calibration paragraphs for fit01–fit05 are independent random samples of 100 WikiText-103 training paragraphs (at least 600 characters; seeds 2026090701–05); the realized paragraph sets are pairwise disjoint and article overlap is 0–3 pairs per fit pair. **fit01** was preselected for confirmation before any new-item outcome; fit02 was retained as a secondary comparison. Fitting ran on an RTX 5090 with an 8-direction CUDA-graph engine whose output was verified bit for bit against the released fitter. Appendix C and the accompanying methods ledger give every setting with its source.

Two further comparisons examine estimator specification. The **target control** changes the derivative target from the final block (virtual 191) to the penultimate block (190) while retaining final normalization and unembedding. The **position controls** place the cotangent at one uniformly drawn valid position q per paragraph and keep all 2,048 derivative directions: *sampled-sum* sums the gradient over all valid source positions (its expectation over q equals the dense estimator), whereas *diagonal* keeps only the source position $p = q$. These are distinct estimator constructions, all fitted on fit01’s calibration paragraphs.

2.2 Prompt, readout position and what counts as recovery

Each question is one unfinished “Fact:” statement ending in a space; the intermediate concept is implied by the clue and never named, and the answer is the continuation. The model input is BOS

plus the prompt tokens up to the boundary where the prompt’s tokenization stops agreeing with the tokenization of prompt-plus-answer (in 131 of 160 items the trailing-space token merges into the answer’s first token and is dropped); the readout position is the last input token, so every input token is causally visible to the scored state. There is no instruction, chat template, demonstration, prefilled response or padding; inputs are 13–35 tokens. Example (confirmation-001): “Fact: The planet whose sidereal rotation takes longer than its orbital period has an atmosphere dominated by” → readout at ” by” (token 20), intermediate **Venus** (scored token 16293, ” Venus”), answer “carbon dioxide”, never shown.

A concept is recovered when at least one accepted single-token form (the name with and without a leading space, capitalized or lower-cased, 1–4 tokens per name) ranks among the first ten tokens of the **full vocabulary**. Each item’s controls are the other 79 intermediate names, equally weighted. Let $H_{i,r,\ell}^m(a)$ be the hit indicator for name a under method m , and a_i the intended name. With control set C_i , the fixed-layer excess score is

$$e_{i,r,\ell}^m = H_{i,r,\ell}^m(a_i) - \frac{1}{79} \sum_{c \in C_i} H_{i,r,\ell}^m(c).$$

The primary endpoint is the equal-weight item mean of the paired difference, averaged across the twelve fixed layers of the fourth pass:

$$\widehat{\Delta} = \frac{1}{160} \sum_{i=1}^{160} \frac{1}{12} \sum_{\ell=26}^{37} (e_{i,4,\ell}^J - e_{i,4,\ell}^{\text{raw}}).$$

This averages fixed-layer measurements; it does **not** ask whether a concept appears anywhere in the band. The historical *any-layer* score separately gives each intended name and each control 48 opportunities per pass, then averages controls, labels within items, and items; it is reported separately.

All effects below are in percentage points. Recovery rates are averages over item–layer cells, not the fraction of questions the model answers correctly. Subtracting control recovery makes the comparison more informative, but does not turn it into a causal test or a calibrated measure of semantic correctness.

3. From discovery to a frozen new-question test

3.1 What the discovery work established

The discovery evaluation contained 90 eligible multihop items (100 labels) and 51 arithmetic items from the released lens-eval sets. Fixed-layer analysis showed that the early multihop loss was not merely an artifact of allowing the raw lens to recover a concept at a favourable late layer: the disadvantage remained in the middle of the stack. Arithmetic was mixed.

Pass 4 differed. Across five independent calibration fits, the mean excess advantage in layers 26–37 was **18.72 points** (between-fit SD **0.39**), positive in every fit, and every fit peaked at physical layer 32 (Figure 1). The final third (layers 33–48) averaged 6.13 points (SD 0.21). The gain predominantly reflected increased intended-concept recovery (18.91 own-name minus 0.19 control points).

That was evidence about calibration variability on the **same questions**, not an independent replication across evaluation populations. The band’s crossed fit/item pointwise interval was 10.96–26.99 points, but its simultaneous interval over the 56 prespecified metrics (−0.86 to 38.29) and its shared-concept interval (−7.49 to 44.92) contained zero, and no positive individual layer cleared the simultaneous shared-concept band. The band was therefore treated as a discovery to test, not a settled advantage.

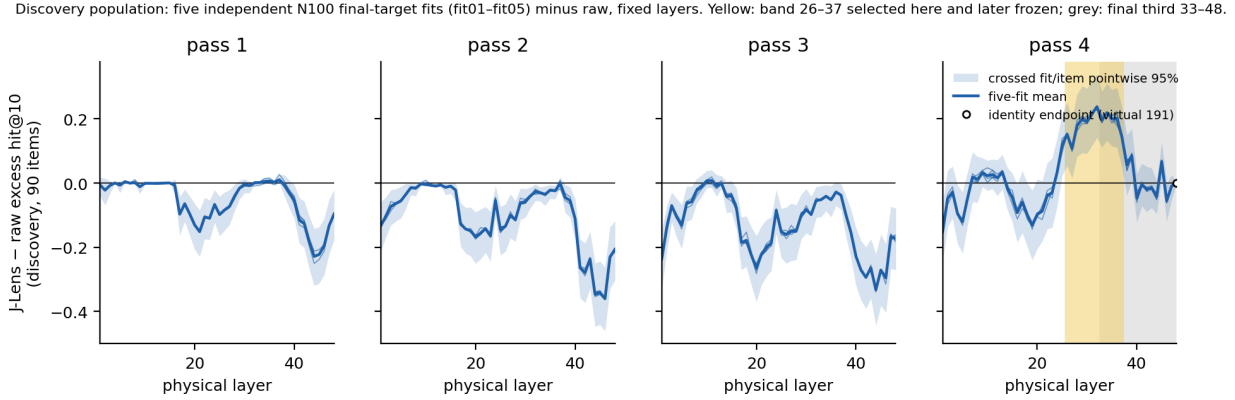


Figure 1: Discovery population: fit01–fit05 minus raw, fixed-layer excess hit@10 for the 90 eligible multihop items, all four passes. Thin lines are the five fits, the bold line their mean, shading the crossed fit/item pointwise 95% interval (not simultaneous). Yellow: layers 26–37, selected here and later frozen; grey: final third. The open point at layer 48 is the identity endpoint. Source: refit analysis arrays.

3.2 Confirmation population and inference

The confirmation set contains **160 two-hop questions, 80 intermediate concepts, two questions per concept**, in ten domains of eight concepts (arts and music, astronomy, biology, chemistry, countries, environment and geology, historical people, landforms, SI units, technology). All 160 were eligible; none of the 80 intermediates or their aliases appears in the discovery set. Construction and annotation were blind to method outcomes; factual hops, novelty and control hygiene were reviewed independently before the local freeze.

The relation mixture differs from discovery: 96 questions request relation types absent from the discovery inventory, 45 use familiar requested relations, 19 recombine familiar relations. This is a transfer test, not a difficulty-matched replication. Three component facts overlap known calibration text (items 076, 130, 151). No claim is made that the facts were absent from pretraining.

Fit01, the pass-4 band, scoring rules, population and analysis were fixed before any new-item outcome; the stopping rule was one evaluation of that population. “Prospective” refers to that recorded local freeze (9 September 2026, 22:40:58 UTC); a public preregistration does not exist. Two runtime corrections frozen before scoring and the later numerical checks are described in Appendix B.

The primary interval uses **20,000 bootstrap resamples of whole dependency groups** (seed 2026090901), with the 2.5th and 97.5th percentiles as limits; resampled groups contribute all their items and the ratio of sums preserves equal item weights. Twenty prespecified secondary contrasts form a separate simultaneous 95% family (bootstrap max- t). Component intervals are descriptive.

Dependency groups. Items are joined into a group when they share an intermediate (or its aliases), an underlying fact, a concrete entity-substitution template (the same typed clue $\rightarrow$ entity $\rightarrow$ requested-slot construction, such as element clue $\rightarrow$ element $\rightarrow$ atomic number), or a normalized final answer or alias code. The 28 connected components have sizes 42, 24, 16, 10, 10, 6, four of 4 and eighteen of 2; the weight-based effective group count is $160^2 / \sum_g n_g^2 = 8.63$. The largest group joins the 16 chemistry items (element $\rightarrow$ atomic-number and element $\rightarrow$ symbol templates) and the 16 SI-unit items (unit $\rightarrow$ symbol and unit $\rightarrow$ quantity templates) through tungsten’s and Watt’s shared answer “W”, with three small clusters (Alps/Andes/Borneo; nucleus; ozone) attached by the small-integer answers “2” and “3”. Leaving out any single group keeps the estimate between 19.73 and 24.87 points (low end without the 16 country items, high end without the 42-item group). The effective count is a weight summary, not a literal number of independent observations, and the grouping does not certify the absence of other dependence.

4. Results on new questions

4.1 The fixed final-pass advantage transfers

In pass 4, layers 26–37, fit01 exceeds the raw lens by **23.19 points** in excess hit@10, with a 95% dependency-group interval of **16.29–32.90 points** (Figure 2, Table 1).

Quantity	Estimate	95% group-bootstrap interval
J-Lens intended recovery	37.92%	30.44–46.56%
Raw intended recovery	14.43%	10.45–17.87%
J-Lens control recovery	0.34%	0.18–0.51%
Raw control recovery	0.04%	0.02–0.08%
Paired excess difference	+23.19 points	+16.29 to +32.90 points

Table 1. Confirmation results for the frozen pass-4 band. Component intervals are descriptive; the final row is the primary inference. Rounded entries need not subtract exactly.

The intended-recovery difference is **23.49 points** (16.62–33.16, descriptive). Control recovery is slightly higher under J-Lens (+0.30 points), reducing rather than creating its advantage: the excess comes from recovering more intended concepts. Resampling individual questions instead gives 17.97–28.71 points.

Fit02 gives a secondary advantage of **22.56 points** (simultaneous interval 9.52–35.59); fit01 minus fit02 is 0.63 points (−0.73 to +1.99). The two fixed estimators perform similarly here; this is not a formal equivalence result. Discovery and confirmation results are not pooled.

Confirmation population (160 new items): fixed-layer curves. Yellow: the frozen primary band, pass 4 layers 26–37. Curves are descriptive; only the band mean was inferential.

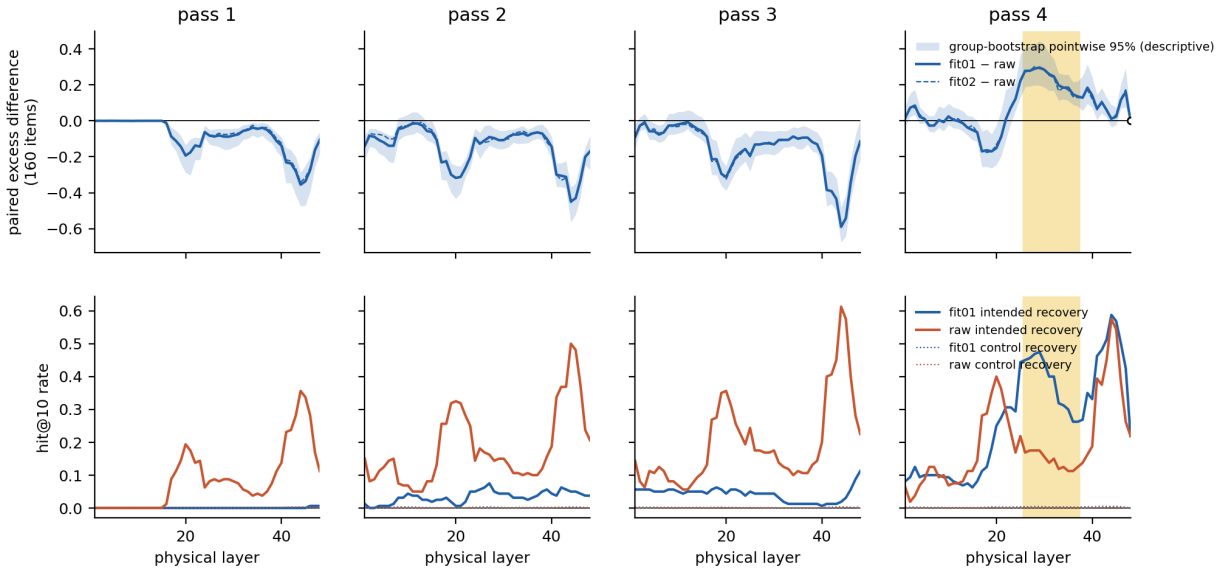


Figure 2: Confirmation population: fixed-layer curves for all four passes. Top: fit01 – raw (solid) and fit02 – raw (dashed) paired excess difference per layer, with the descriptive group-bootstrap pointwise 95% band for fit01. Bottom: intended recovery for fit01 and raw, with control recovery dotted. Yellow: the frozen primary band. Only the band mean carries the prospective inference. Source: accepted payload readouts.

4.2 The early-pass deficit transfers, including in the same physical band

All six prespecified early-pass comparisons remain negative on the new population (Table 2).

Pass	Mean over fixed layers 1–48	Any-layer recovery
1	−8.83 [−13.64, −4.01]	−44.41 [−59.03, −29.78]
2	−14.86 [−19.54, −10.18]	−57.37 [−72.01, −42.72]
3	−15.59 [−22.40, −8.77]	−61.08 [−79.59, −42.58]

Table 2. *Fit01 minus raw excess hit@10, in percentage points. Brackets are simultaneous 95% intervals from the 20-contrast secondary family.*

These summaries alone do not show what happens in the specific band that carries the pass-4 gain. As a **post-confirmation reviewer analysis**, the same physical layers 26–37 were evaluated in every pass with the frozen estimator, controls and groups (Table 3, Figure 3). The band shows a deficit in each early pass whose simultaneous interval (a four-contrast family, Appendix B) excludes zero; in pass 1 fit01 recovers no intended label in the band at all. The pattern is therefore pass-dependent at fixed physical depth, not an artifact of averaging over 48 layers.

Pass	J-Lens intended	Raw intended	Paired excess difference	Interval	Status
1	0.00%	6.46%	−6.50	[−11.91, −1.08]	post-confirmation, simultaneous
2	4.22%	12.71%	−8.64	[−12.43, −4.85]	post-confirmation, simultaneous
3	2.76%	13.91%	−11.14	[−16.76, −5.52]	post-confirmation, simultaneous
4	37.92%	14.43%	+23.19	[+16.29, +32.90]	prospective primary

Table 3. *The same physical band, layers 26–37, in every pass. Control recovery is below 0.35% in every cell.*

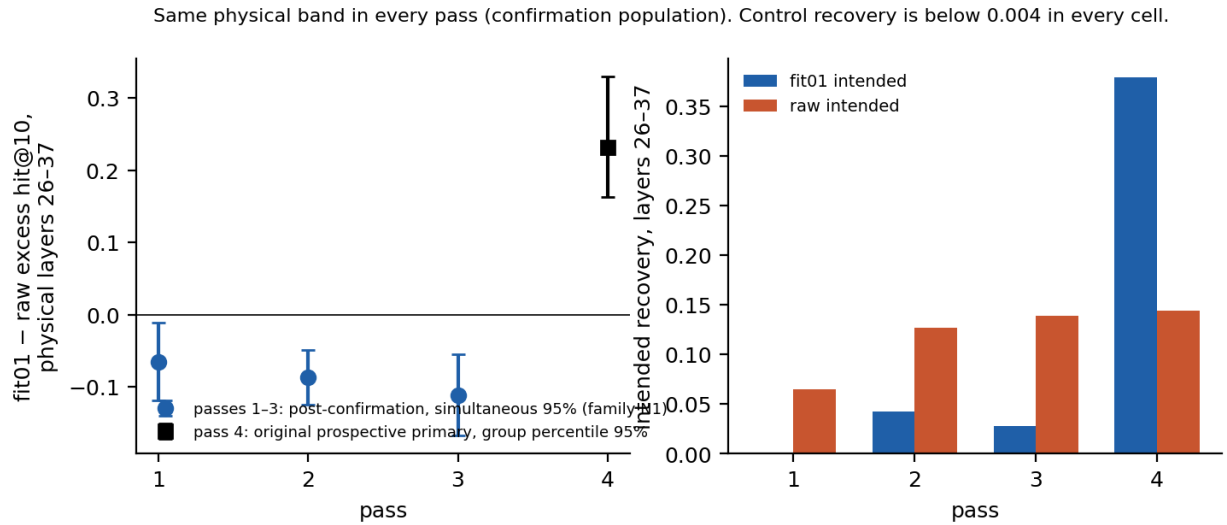


Figure 3: Layers 26–37 in every pass on the confirmation population. Left: paired fit01 – raw excess difference; passes 1–3 carry the post-confirmation simultaneous 95% interval, pass 4 the original prospective percentile interval. Right: intended recovery by method.

4.3 The estimator specification matters

The control comparisons transfer to the new questions (Table 4, Figure 4).

Contrast in pass 4, layers 26–37	Difference	Simultaneous 95% interval
Main fit01 minus penultimate target	+14.38	[+7.70, +21.07]
Sampled-sum minus diagonal	+15.77	[+5.80, +25.75]
Penultimate target minus raw	+8.80	[+0.76, +16.85]
Sampled-sum minus raw	+21.72	[+9.16, +34.29]
Diagonal minus raw	+5.95	[−0.34, +12.24]

Table 4. Estimator controls in percentage points; matched layer support within each control family; same 20-contrast family as Table 2.

Changing the derivative target by one block reduces the advantage substantially; restricting the position reduction to the matching source position does so as well. Neither comparison isolates how Ouro reasons: it shows that the measured recovery depends on the lens’s construction. The diagonal-only estimate is smaller and its support is borderline: the lower interval bound is within 0.4 points of zero on every numerical path and excludes zero only under regenerated states. The direct sampled-sum-minus-diagonal comparison is the clearer result.

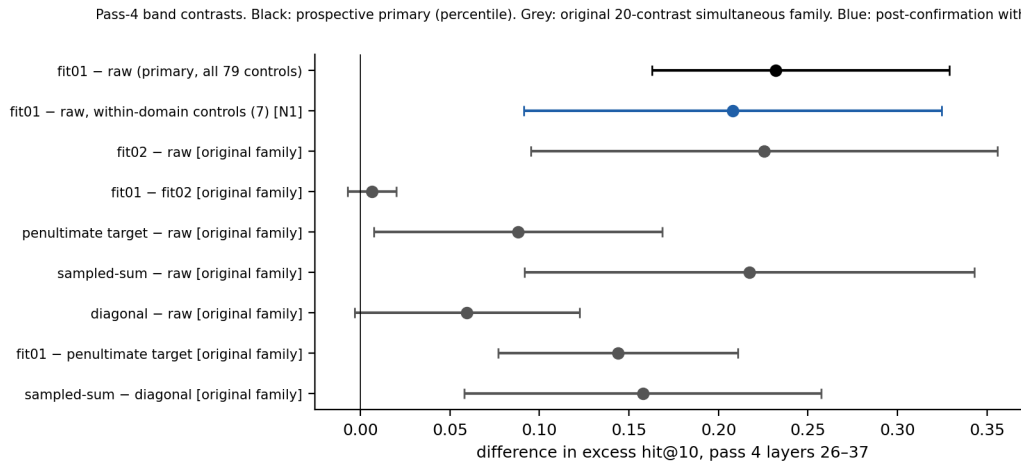


Figure 4: Pass-4 band contrasts. Black: the prospective primary (percentile interval). Grey: contrasts of the original 20-member simultaneous family. Blue: the post-confirmation within-domain-control contrast (Section 4.4).

4.4 Post-confirmation reviewer checks

Three further analyses were specified on the already-used confirmation data after the confirmation; they are reported separately and do not alter the primary endpoint.

Within-domain controls. Replacing each item’s 79 controls by the seven other names in its benchmark domain gives a paired excess difference of **20.80 points** (simultaneous 95% interval 9.13–32.46). J-Lens recovers same-domain names at 3.07% against 0.34% for arbitrary controls (raw: 0.38% against 0.04%), so about a ninth of the lens’s gain is domain- or type-level rather than concept-specific. An arithmetic bound derived from the published values (intended difference minus 79/7 times the J-Lens all-control rate, 19.61 points) is satisfied by the exact arrays for every item. Four of the ten domains contain a single semantic type (elements, countries, SI units, people);

the others mix types (planets and moons; rivers, ranges, a desert and an island; molecules, a cell type and an organelle), so this is a within-domain control, not a same-type control. The gain is heterogeneous across domains: from about +51 points for countries and +40 for environment and geology to about zero for SI units, where neither method recovers unit names in this band, and to −1 for astronomy (+5 with the 79 controls), where the lens often recovers other planets and moons. Domains are not independence units.

Tokenized-input overlap. For every item, every scored token of the intended label and of all 79 controls was compared with the exact model input. No intended-label token appears in any input. Six control tokens do, in six items: “Voyager” in three astronomy prompts, “Chile” in an Andes prompt, “Egypt” in a Nile prompt, and the lower-case form “io” as a token inside “Fidelio”. Dropping those controls for the affected items (both methods; 78 controls for six items) gives 23.19 points (16.25–32.72); dropping the six items gives 22.90 points (15.92–33.15). Lexical word-level mentions without scored-token overlap occur once more (“newton” in a pascal item). Zero token overlap does not establish absence of semantic clues or of the facts from pretraining; the prompts are designed to imply the intermediate.

Grouping. The rule, memberships and the composition of the 42-item group are given in Section 3.2 and the reviewer ledger; no alternative grouping was constructed.

5. Numerical verification

The confirmation run used BF16 model execution on an RTX 5090. A development check initially failed because applying the output head to one residual row did not reproduce logits computed in a multi-row call. The subsequent verification measured whether this affected the scored comparisons rather than assuming a small logit discrepancy was harmless. These checks were **post-confirmation**, not prospectively registered.

On a local RTX 5070 Ti with the confirmation’s torch build and precision settings, the retained confirmation states, verified banks and frozen scoring code reproduced the saved arrays bit for bit across **160 items, six arms and all scored columns**, including ranks, top-ten token IDs, saved exit logits and sampled transported vectors. Calls with 160, 190, 191 and 192 rows matched the executed layout exactly; single-row calls changed 1.5–1.6 million of about 1.5 billion logits per arm by at most 0.125 (0.25 for raw) but no endpoint quantity.

Computation	Primary difference	95% group interval	Change from accepted
Accepted BF16 path	+23.19	[+16.29, +32.90]	—
Alternative row packing	+23.19	unchanged	0.00
FP64 reference computation	+22.93	[+16.17, +32.52]	−0.26
Locally regenerated states	+23.61	[+16.52, +33.77]	+0.42

Table 5. Numerical sensitivity of the primary endpoint. The FP64 row applies a reference readout computation to the stored tensors, not an FP64 model. Alternative packing was measured on the local GPU, not in a fresh RTX 5090 run.

CPU and GPU FP64 computations gave identical ranks, hits and top-ten sets (values differing by at most 9×10^{-14}). Resolving every exactly tied logit adversarially bounds the accepted estimate between **22.60 and 23.35 points**; moving every logit by up to 1/32 or 1/8 in the adversarial direction gives 22.01–24.41 and 18.30–27.26. Regenerating states from the prompts on the local GPU was not bit-identical to the RTX 5090 forward: 37 of 160 state records matched exactly and 154 greedy continuations matched, yet replaying every arm on those states moved the endpoint by

only 0.42 points. All six early-pass contrasts and the direct estimator-control contrasts keep their signs and zero-exclusion on every path. Bit-level state regeneration on another RTX 5090 was not tested.

A separately implemented checker (a separate coding agent working from a specification, without producer code) reconstructed the saved analysis to within 9×10^{-16} and detected all 13 planted errors in isolated fixtures. “Independent” here means a separate implementation, not replication by another laboratory.

6. Local-exit targets: a conditional comparison, partly blocked

The main bank targets the final pass, so early-pass states are far from its target: a pass-1 state at layer 30 is 162 blocks and three pass-boundary normalizations away from virtual layer 191. One explanation of the early deficit is therefore transport distance rather than a property of early-pass representations. The intended test evaluates the same stored early-pass confirmation states with the raw lens, a J-Lens targeting the current pass’s own exit, and the final-target J-Lens, in layers 26–37 of passes 1–3.

Inventory. Local-exit banks exist only from the application era (5 September 2026): banks targeting virtual 47, 95, 143 and 191, fitted on the first 100 paragraphs of the historical calibration file (12 articles) with one shared recipe that differs only in the target. Their merged files were purged; the final-target bank and four of its five shards survive locally, whereas the twelve shards of the three local-exit banks survive only in a private repository (hashes recorded, no credential on this machine). The confirmation-population comparison is therefore **blocked**, and no comparison was improvised. Retrieval alone (11 GB, no fitting, no paid compute) would unblock it; a fit01-matched refit would cost about eight GPU-hours and would be a new estimator family.

What the retained readouts show (descriptive, discovery population). The application-era banks were all applied to the 148 discovery items in one evaluation whose rank arrays survive. On the 90 eligible multihop items, in the fixed band 26–37, the lens targeting each pass’s own exit recovers the intended concept far more often than both alternatives: intended recovery 15.4%, 16.2% and 25.2% in passes 1–3, against 2.1%, 4.3% and 10.1% for the raw lens and 0.3%, 0.4% and 2.1% for the final-target lens (Figure 5). The paired excess differences, local minus final, are +14.6, +16.9 and +23.0 points (shared-concept-cluster percentile intervals 9.1–20.9, 11.5–22.8 and 16.0–30.6), and local minus raw +13.0, +11.3 and +14.9 points (8.1–19.0, 6.6–16.6, 7.9–22.4). Control recovery stays below 2.3% everywhere.

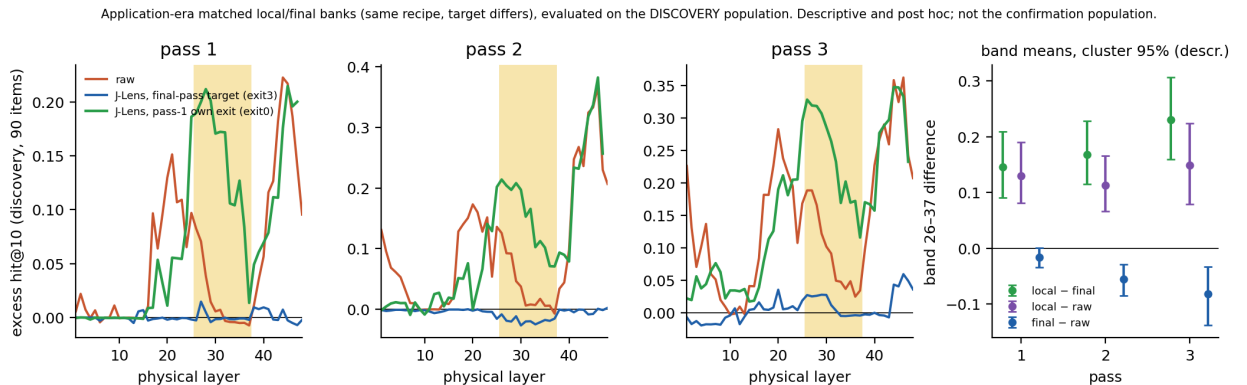


Figure 5: Application-era matched banks (same recipe, target differs) evaluated on the DISCOVERY population, fixed layers, passes 1–3: raw lens, final-target J-Lens (exit 3) and the pass’s own-exit J-Lens. Right: band means with descriptive shared-concept-cluster 95% intervals. Post hoc and descriptive; not the confirmation population.

Interpretation. Within the rules fixed before these numbers were computed, this is the case in

which the local target beats both alternatives: a target-dependent early-pass advantage, consistent with a transport explanation. It is not proof that averaging destroyed information; it is on the population on which the band was selected; it uses the older 12-article calibration and a different batch setting; the intervals are descriptive; and it has not been tested on the confirmation population. Two existing qualifications remain: shortening the final target by one block (the penultimate control) reduced the late gain, so transport distance alone is not established, and raw pass counts are not comparable transport lengths across architectures. The historical Huginn pilot (Appendix A) did not identify a cause or rule out supervision-based explanations.

7. Interpretation and limits

The supported result is a **localized improvement in reading annotated intermediate concepts from fixed Ouro states**: a specific final-pass band, one frozen estimator, transferred to new questions and concepts without refitting or band selection. The same band in the earlier passes shows the opposite sign, so the method is not uniformly better as computation proceeds, and the source of the measured gain is higher intended-concept recovery, part of which is domain-level.

One possible explanation for the strong raw baseline is Ouro’s training objective, which applies weighted next-token losses at recurrent pass boundaries through a shared output head [2]. This does not supervise every physical block, and the present experiments do not isolate the effect of that objective. The target-dependence seen on the discovery population points to transport distance or target choice as a contributing factor, but architecture, calibration, target choice and executed depth remain entangled, and the Huginn pilot’s evidence is limited by its incomplete artifact chain.

Recovery is not use. In exploratory checks a lens recovered a labelled intermediate on questions the model answered incorrectly, and native exits can agree while their readouts disagree sharply (Appendix A).

The main population limit is the small number of unequal dependency groups. The interval accounts for the declared grouping, not every possible shared fact, relation or template; the effective count of 8.63 warns against reading 160 related questions as 160 demonstrations. The relation mixture is not difficulty-matched, labels are restricted to accepted single-token forms, the domain heterogeneity is large, and the evidence concerns one checkpoint and two retained calibration fits. The study does not establish performance on arbitrary concepts, other models, or monitoring under adversarial pressure.

Reproducibility and artifact availability

The confirmation banks, states, scores, frozen inputs and code permit exact table reconstruction, rescoring from banks and states, and forward replay (not bit-identical off the RTX 5090). Complete refitting of the original estimators is not available: an earlier retrieval failure and a later storage purge destroyed unique historical artifacts, including all four Ouro FP32 fitting checkpoints, the historical discovery-evaluation state cache, the truncated Huginn checkpoint and the application-era local-exit banks (43 purged paths, 33 distinct contents lost; 19 paths unresolved in a remote repository). Knowing the recipe does not restore those checkpoints, and the Huginn pilot is retained only at the saved-output level. On 11 September 2026 a copy-only archive of every surviving file (16,200 files, 52.15 GB, every byte re-hashed on the destination) was written to one external SSD and restoration-tested from that SSD alone with the laptop’s paths hidden and no network: the frozen code matched its freeze, all four bank files loaded, the frozen analysis reproduced the primary and all twenty secondary contrasts exactly, and raw and fit01 rescoring from the archived states and weights reproduced the saved ranks bit for bit. Until a second destination exists it is a single-copy archive, not a redundant backup. The frozen records, code, compact results, figures and this manuscript are published in the repository <https://github.com/VykosMolt/JLens-Ouro> (research directory; files of 50 MB or more, including the banks and state caches, are listed there

with hashes but not stored). No DOI exists at the time of writing.

8. Conclusion

J-Lens loses to raw readout in Ouro’s first three passes, including in the very layers where it wins in the fourth, and recovers more annotated intermediate content in a fixed region of the final pass. That advantage transferred to new questions and concepts and survives the measured numerical variations and control redefinitions. Its size depends strongly on the derivative target and the position aggregation, and retained discovery-set readouts suggest that a lens targeting the pass’s own exit reads early passes far better than the final-target lens. The finding is about where a particular readout works, not about general superiority, successful reasoning or causal control.

Appendix A. Exploratory findings kept separate from confirmation

Exit disagreement

In the 7 September analysis of 148 prompts, native pass-3 and final predictions agreed **85.1%** of the time (item-bootstrap interval 79.1–90.5%). Two J-Lens variants targeting the current and final exits agreed only **1.45%** (0.58–2.51%) averaged over physical layers 41–47; layer 48 was excluded because some agreement there follows by construction. One token earlier, native agreement rose to 96.6% while lens agreement remained 0.39%. These are different comparisons, so the lens gap cannot be read as the amount of model revision, and poor output prediction does not by itself refute an intermediate-recovery method [1].

Passing continuations and arithmetic

The early multihop deficit remained when the discovery items were separated by the historical continuation-pass criterion. That criterion was imperfect: two numerical false positives reduced the original 72 passes out of 148 to 70, while additionally accepting the numeric equivalent of one word-form target gave 71. Four-token continuations and lexical matching do not establish semantic correctness; the subgroup observations are exploratory.

A separate supervised arithmetic probe scored 59.4% against the raw lens’s 76.4% in pass 1, but exceeded it in pass 3 (35.6% versus 17.9%). The classifiers trained on only 36 independent operand pairs; a bounded training-size comparison (+3.9 points, interval -2.9 to $+10.9$) was inconclusive. This used a different task family and a candidate-set metric, with incomplete provenance for the older J-Lens comparator; it is not pooled with the concept-recovery results.

Huginn pilot

The pilot used one Huginn estimator calibrated on the same 100 texts as the preselected Ouro fit, eight recurrent passes and two evaluation initializations. Joint tokenizer eligibility retained 88 multihop items (98 labels) and 51 arithmetic items. J-Lens-minus-raw estimates were negative under all six prespecified summaries in both initializations; the cross-model differences of differences had mixed signs and simultaneous intervals containing zero. The prediction of a larger relative J-Lens advantage in Huginn was not supported. This does not isolate supervision: Huginn’s coda is trained after sampled recurrence depths [3], so it is not an unsupervised-intermediate control, and its raw-readout weakness is block-specific [4]. The fitted Huginn bank was never retrieved and its partial checkpoint was later lost; the saved statistics are historical evidence, not a verified cross-model replication.

Appendix B. Verification and analysis provenance

The original confirmation freeze is unchanged. Two separately recorded corrections preceded new-item scoring: `precision_v1` (10 September, 10:44:52 UTC) removed redundant TF32 setters, and `native_check_v1` (22:56:18 UTC) changed the development comparison from a single residual row to a whole-sequence readout and resized run budgets. The checker’s accepted run-specification pin was updated after outcomes were known. The post-confirmation verification reviewed these changes and found the scientific input, estimator, endpoint and analysis unchanged.

The numerical results use the accepted BF16 endpoint as primary; FP64, alternate packing, tie handling and regenerated states are sensitivity checks. For diagonal minus raw, the accepted estimate is +5.95 points with simultaneous interval -0.34 to $+12.24$; FP64 gives -0.40 as lower bound and regenerated states $+0.04$, which is why its support is described as borderline.

Post-confirmation reviewer family (Section 4.2 and 4.4). Exactly four new inferential contrasts were fixed in a dated plan before computation: the within-domain paired excess difference in the pass-4 band and the same-band paired differences in passes 1–3. They share one centered bootstrap max- t 95% family (20,000 whole-group draws, seed 2026091101, quantile 2.639). Every other new quantity is descriptive. The overlap sensitivities and the local-exit discovery-population summary are descriptive.

Records. Confirmation run `ouro_confirmation_20260911_fixed160_native1`; frozen plan, benchmark and freeze receipt under `research/confirmation_2026-09-09/`; verification under `research/verification_2026-09-11/` (FINAL_VERIFICATION.md, REPORT_confirmation_verified.md, RUN_SPECIFICATION.json); this round’s reviewer checks, methods ledger, numerical source ledger and archive receipt under `research/closeout_2026-09-11/`. These are project provenance identifiers, not public links.

Appendix C. Method details in brief

- **Tokenization and input.** BOS + prompt tokens (`truncation=True`, `max_length=511`, never triggered); readout context = common prefix of `encode(prompt)` and `encode(prompt+answer)`; position -1 ; batch size 1; SDPA attention; `use_cache=False`; BF16 weights.
- **Hooks and normalization.** Forward hook on each shared physical block filtered by the recurrent step; recorded tensor = post-residual block output, before the pass-end RMS normalization; states stored as FP32 copies of BF16 values.
- **Readout.** Jh in FP32 (raw: h), cast to BF16, RMS-normalize (FP32 internally, $\epsilon = 10^{-6}$), BF16 head, one call per item over all scored columns; descending argsort over 49,152 tokens; rank = min over accepted forms; hit = rank < 10 .
- **Calibration and fitting.** 100 WikiText-103 paragraphs per fit (fit01–05 seeds 2026090701–05; paragraph-disjoint; 100/99/99/100/100 distinct articles), inputs of 128 tokens including BOS (six shorter), valid positions 16 to $\text{len}-2$, cotangent summed over valid target positions and gradient averaged over valid source positions, FP32 sum over paragraphs divided by 100, FP16 storage, 8 output directions per backward pass with CUDA-graph replay (bit-parity verified), RTX 5090.
- **Controls.** Penultimate: target virtual 190, sources 0–189. Positions: one uniform q per paragraph (seed 2026090801), sampled-sum = $\sum_p \partial h_{191}[q]/\partial h_v[p]$ (no divisor), diagonal = $\partial h_{191}[q]/\partial h_v[q]$, from one backward stream. Matched supports for contrasts: band 26–37 for all arms; final-third intersections 176–189 (target control) and 176–190 (position controls).
- **Banks.** fit01 90f01f6a..., fit02 101f31db... (byte-exact reconstruction from its intact checkpoint), penultimate dc6354df..., positions b8e8b7d2... (both position arms); four files, five bank arms, six evaluated arms.
- **Application-era family (Section 6).** Same estimator with the first 100 historical paragraphs (12 articles), 32 directions per backward pass, B300 pod; targets 47, 95, 143, 191.

Appendix D. Contribution and AI-assistance statement (draft)

AI coding agents implemented the fitting, evaluation, controller and analysis code, executed the runs under the author’s authorization, and performed separately instantiated review and numerical-reconstruction passes; the project materials name Claude Code, Claude Opus and Codex for the application era, and round logs after 7 September do not record agent versions. The author set the research questions, approved the frozen design and budgets, authorized each paid run and the storage purge, and is responsible for all claims. Both manuscript drafts were prepared by AI assistants from the frozen reports and records; the extent of the author’s verification of each number is to be stated by the author. No venue policy has been assumed. [Items requiring the author’s confirmation are listed in the accompanying fact sheet.]

References

- [1] Gurnee, W., Sofroniew, N., Pearce, A., et al. *Verbalizable Representations Form a Global Workspace in Language Models*. Transformer Circuits, Anthropic, 6 July 2026. Methods; Quantitative Methodological Comparisons; Methodological Details and Ablations. <https://transformer-circuits.pub/2026/workspace/>
- [2] Zhu, R.-J., Wang, Z., Hua, K., et al. *Scaling Latent Reasoning via Looped Language Models*. arXiv:2510.25741, 2025; Sections 3.1–3.3 for the architecture and training objective. <https://arxiv.org/abs/2510.25741>
- [3] Geiping, J., et al. *Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach*. arXiv:2502.05171, 2025; Sections 3.1–3.3 and 6.1 for the Huginn architecture, training and early exits. <https://arxiv.org/abs/2502.05171>
- [4] Lu, W., et al. *Latent Chain-of-Thought? Decoding the Depth-Recurrent Transformer*. arXiv:2507.02199, 2025; Section 3.2 for block-specific logit-lens readability in Huginn. <https://arxiv.org/abs/2507.02199>